

Introduction

This initial work evaluates how reservoir computers fit into computational applications between edge and agent devices by exploring the generalizable accuracy across time series prediction tasks.

Time series computation is an important part of decision making processes in autonomous and self-aware computational systems.

Reservoir computers are lightweight, space efficient, computationally efficient, and quick to train in contrast to transformers and other neural network models.

Research into computational applications between edge and terminal devices on the network has grown commensurate with the growth of ubiquitous computation and automation.

Methodology

Data generation and training of the readout layers and the reservoir state updates are executed on separate ROS 2 nodes each housed in separate docker containers on a simulated network.

The system is constructed from two primary ROS 2 nodes:

Agent Node

- Generates time-series data
- Publishes batches of state values to the network
- Receives processed reservoir outputs
- Trains output layer

Edge Node

- Implements the reservoir
- Receives state data from the agent
- Computes reservoir state evolution using a fixed recurrent weight matrix
- Transmits the updated reservoir state back to the agent

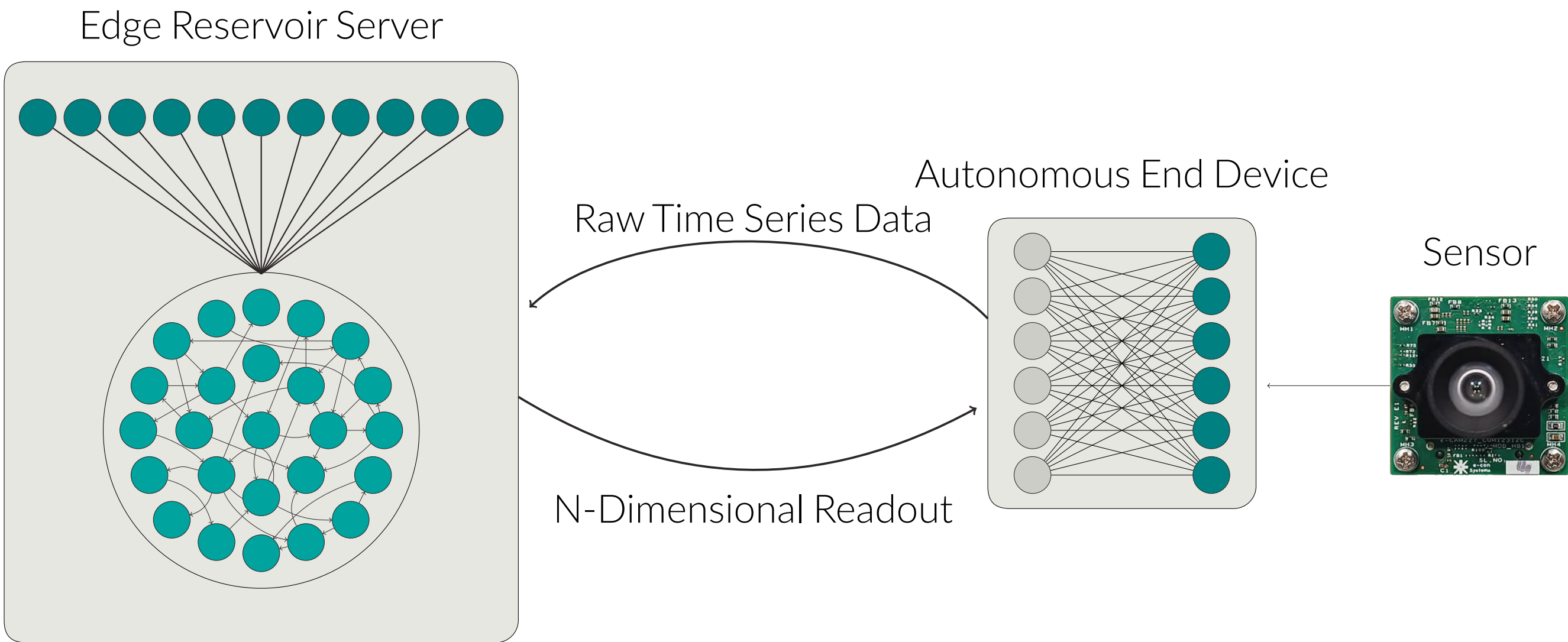


Figure 1. A graphical representation of a distributed reservoir computer.

Reservoir Configuration

The reservoir implementation used in this project follows the classical Echo State Network formulation [1, 2, 3].

We choose the following hyperparameters for our reservoir:

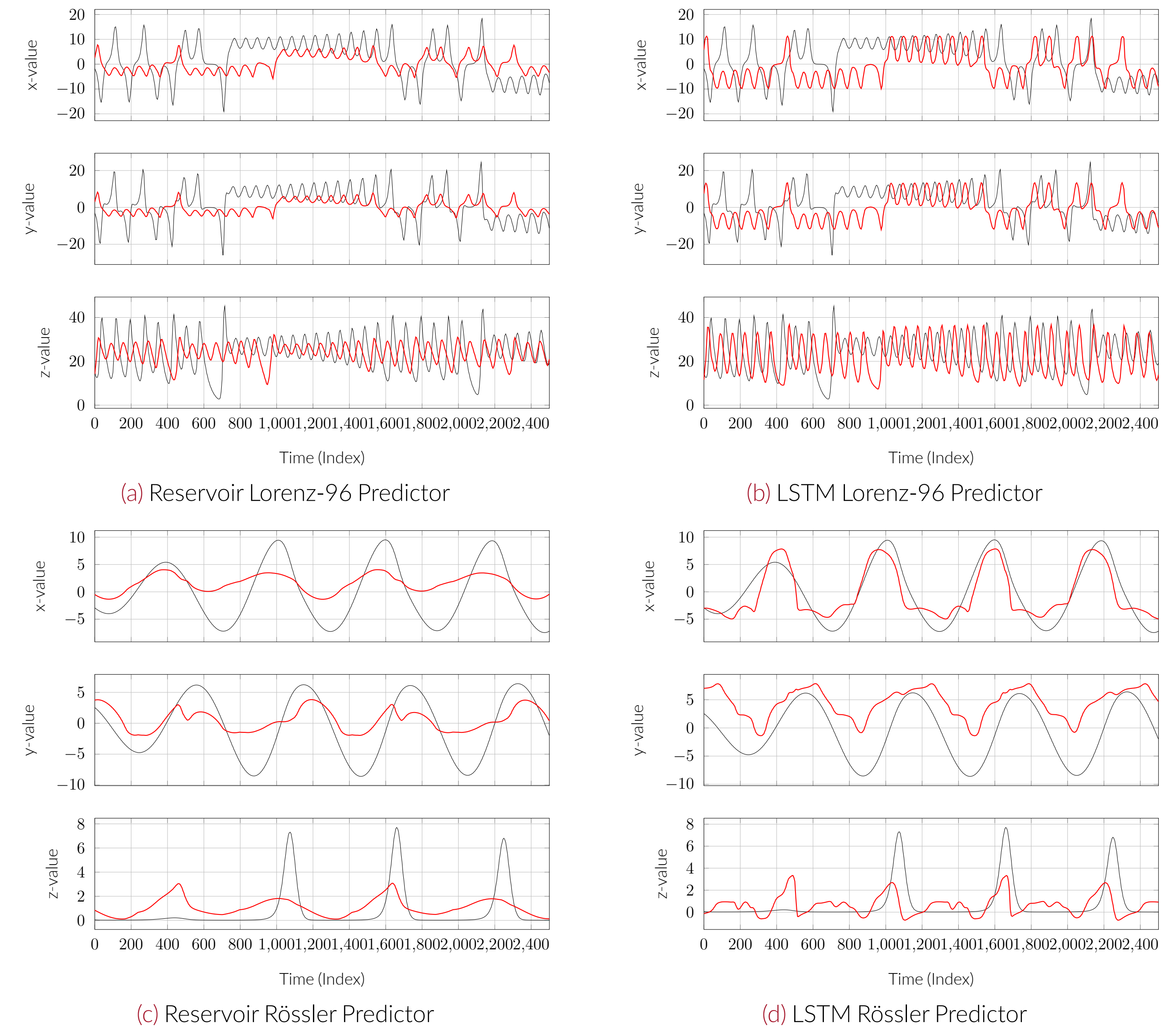
- Reservoir dimension is set to 1024.
- Leak rate is set to 0.1.
- Spectral radius of the reservoir is set to 1.1.
- Reservoir is instantiated with the same seed and random number generation algorithm.
- Weight matrix is initialized using a power-law distribution to create a spectrum of weights with a heavy-tailed structure.
- The readout layer is trained using a linear ridge optimization.

Results

Figure 2 plots the results of 2500 time step predictions of the Lorenz and Rössler systems by a reservoir predictor with a readout layer trained for the respective system.

Both reservoir predictors have comparable accuracy with a low RMSE.

The reservoir preserves the characteristic chaotic behavior of both systems without exceeding the bounds of either system, and the predictors achieve accuracy on par with state of the art reservoir computer implementations [1, 2].



Discussion

This initial work consisted of the designing and building of the test architecture.

These preliminary results results introduce the utility and efficacy of distributing the layers of a reservoir computer between an edge and agent device in a network and supports our hypothesis that reservoir computer architecture may be distributed over a network across more than one device.

By abstracting the computational layer of the edge node from the surrounding data pipeline, any computational model or framework may be dropped in as a replacement for the reservoir, enabling direct comparison across architectures under identical network and workload conditions.

This makes the system useful not only as a testbed for reservoir computing, but as a general-purpose evaluation platform for emerging edge computing paradigms.

Future Work

We are currently extending the software framework developed in this initial evaluation to dynamically support multiple agents connecting concurrently to the edge reservoir server and further measure the resiliency of reservoir computation under simulated adverse network conditions.

These same dynamical system benchmarks will serve as a standardized baseline for evaluating the generalization of the reservoir’s computational capacity across agents and workloads.

Upon completion, the full framework will be released as open-source software on GitHub.

This framework will then be applied to the computational analysis of real-world, real-time, *n*-dimensional time-series datasets, including:

- Sepsis detection
- Dynamic computational load estimation for simultaneous localization and mapping (SLAM)
- Ambient acoustic noise classification in mobile underwater sensor arrays

These applications are chosen to stress-test the system across domains with fundamentally different temporal dynamics, noise profiles, and latency requirements. We welcome suggestions for additional evaluation domains that may further probe the capacity and limitations of this distributed architecture.

Acknowledgments

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References

- [1] Andrea Ceni and Claudio Gallicchio. Edge of stability echo state networks. August 2023. doi: 10.48550/ARXIV.2308.02902.
- [2] Daniel J. Gauthier, Erik Bollt, Aaron Griffith, and Wendson A. S. Barbosa. Next generation reservoir computing. *Nature Communications*, 12(1), September 2021. ISSN 2041-1723. doi: 10.1038/s41467-021-25801-2.
- [3] Grigoryeva Lyudmila and Juan-Pablo Ortega. Echo state networks are universal. June 2018. doi: 10.48550/ARXIV.1806.00797.